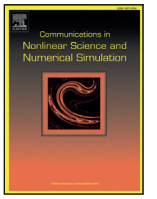
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Research paper

Application of ESN prediction model based on compressed sensing in stock market



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ABSTRACT

Echo State Network (ESN) is often used for time series prediction and chaotic series prediction. But in the training process of ESN model, the reserve pool will involve a large number of calculations, so the reserve pool may have node redundancy, resulting in inaccurate training model. It is very important to select the active nodes in the reserve pool because the predicted nodes are very few compared with all the nodes in the reserve pool. Based on the characteristics of the original vector, compressed sensing can reduce the dimension of the vector so as to obtain the activated nodes. In this paper, the ESN model is added with compressed sensing, which simplifies the reserve pool part of the ESN model. The simulation results show that our method can not only reduce the calculation amount of training effectively, but also improve the accuracy.

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1. Introduction

ESN is a new recursive neural network proposed by Jaeger in 2001 [1], which can avoid the slow convergence speed, high computational complexity and instability of the recursive neural network learning algorithm [2,3]. So ESN becomes a hot issue in academic community, and is widely used in various fields, including dynamic pattern classification, robot control, object tracking and moving target detection, event monitoring, etc. [4,5], especially on the time series prediction problem, made outstanding contributions. And the ESN in time series prediction and chaotic prediction have done a lot of work, but because of the reserve pool of ESN model involves a large amount of calculation, the ESN training and prediction for a long time [6]. ESN also has many algorithms, such as Pseudo inverse method, Wiener-Hopf method, Ridge Regression (Tikhonov regularization), etc [7].

The compressed sensing (CS) proposed by Tao can obtain the discrete samples of the signal under the condition that the sampling rate is much lower than Nyquist sampling rate, so as to ensure the signal reconstruction without distortion [8]. The key idea of compressed sensing theory mainly includes two points. The first is the sparse structure of the signal. The traditional Shannon signal representation method only uses the least priori information of sampled signals, namely the bandwidth of signals. However, many of the signals that attract much attention in real life have some structural characteristics of their own [9–11]. Compared with the DOF(Degree of Freedom) of bandwidth information, these structural

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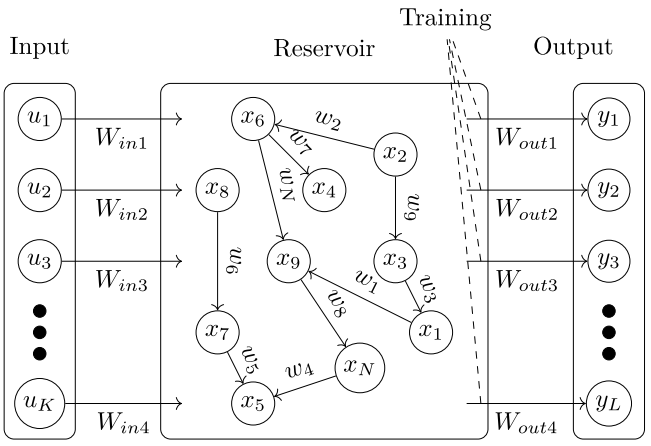


Fig. 1. Where u_i is the input layer node, x_j is the reserve layer node, and y_k is the output layer node($i = 1, 2, \ldots, K$; $j = 1, 2, \ldots, N$; $k = 1, 2, \ldots, L$).

characteristics are determined by a smaller part of the DOF of the signal. The other is the irrelevant properties. The useful information of sparse signals can be obtained by a non-adaptive sampling method to compress the signals into smaller sample data. Because compression has stability (acquisition and reconstruction processes are both numerically stable); generality (the same measurement projection can be used for different signal acquisition); asymmetry (encoding (sampling) is simple, most of the calculation is decoding (reconstruction)); democracy (each measurement carries the same information, which is robust to the loss at the time of measurement). Therefore, the trained ESN reserve pool is reduced in dimension by using compressed sensing to retain only the nodes with the highest value (feature nodes), that is, the activated nodes are screened out and the activated nodes are used for prediction [12,13]. Since only active nodes are used in the prediction process, as long as the model is trained well, the prediction speed will be greatly accelerated. Moreover, the interference of other inactive nodes is relatively small, which can make the error of the whole model smaller, and the mean value and variance value of the error will also be reduced [14].

In this paper, an ESN prediction model based on compressed sensing is proposed to get better prediction results with fewer nodes. The main contributions of this paper are as follows:

1. A new algorithm is proposed to optimize the ESN reserve pool using compressed sensing to improve the prediction speed and accuracy.
2. Analyze the predicted data and compare the advantages and disadvantages of the original ESN model and the ESN model with CS added.
3. Including the use of conventional chaotic systems for simulation verification, we also choose more complex stock data to verify the effectiveness and accuracy of the proposed model. The simulation results are satisfactory.

The rest of this article is as follows. Section 2 describes ESN model and compressed sensing model. Section 3 will introduce the new ESN model with compression sensing, and give its flow chart and algorithm. Section 4 will give two simulations, one is the prediction of Logistic chaotic time series, the other is the prediction of 200 stocks. Finally, we provide a summary of the whole work.

2. Preliminary

2.1. Echo state network

2.1.1. The structure of the ESN

ESN consists of three parts: 1) an input layer; 2) a reserve pool; 3) an output layer. Where, W_{in} connects the input layer and the reserve layer, W reserve layer internal connection, W_{out} connects the reserve layer and the output layer. Fig. 1 shows the basic structure of ESN. W_{in}, W and the reserve pool are initialized without training. We only need to train the connection between the reserve layer and the output layer W_{out}. Since nodes generated by dynamic reservoirs have short-term memory capacity, we can map data from the lower dimension to the higher dimension, and then reduce the higher dimension data to the specified dimension by W_{out}, so as to realize the prediction process.

Since the ESN prediction model needs to raise the dimension through the reserve pool and reducing the dimension, we need to activate the function at both the reserve pool and the output layer, and the updated rules are as follows:

$$\begin{aligned} x^{(n+1)} &= f(W_{in} \cdot u^{(n+1)} + W \cdot x^{(n)}) \\ y^{(n+1)} &= f_{out}(W_{out} \cdot x^{(n+1)}) \end{aligned} \tag{1}$$

Table 1
Commonly used activation functions.

Function	Expression
tanh	$f(x) = \tanh(x)$
identify	$f(x) = x$
softmax	$f(x) = \frac{e^x}{\sum_i v(i)}$
sigmoid	$f(x) = \frac{1}{1+e^{-x}}$
inverse sigmoid	$\ln(\frac{1}{x} - 1)$

where $u = (u_1, u_2, \ldots, u_K)^T; x = (x_1, x_2, \ldots, x_N)^T$ and $y = (y_1, y_2, \ldots, y_L)^T$; in the number field $\mathbb{P}$, $W_{in} \in \mathbb{P}^{N\times K}$is the connection matrix between the input layer and the reserve pool; $W \in \mathbb{P}^{N\times N}$ is the self-connecting matrix of the reserve pool and $W_{out} \in \mathbb{P}^{L\times N}$ is the connection matrix between the reserve pool and the output layer; f is the activation function of the reserve pool and f_{out} is the activation function of the output layer. Commonly used activation functions are shown in Table 1.

In order for the ESN model to have an echo state nature, we need to ensure that the spectral radius of the connection weight matrix inside the reserve pool is less than 1. In other words: $\lambda_{max} = \max\{abs(eig(W))\}$, only when the $\lambda_{max} < 1$, ESN has an echo state attribute. To reduce the spectral radius to α:

1. Calculate the spectral radius α_0 of W;
2. Normalize W by dividing by the spectral radius of W($W := \frac{W}{\alpha_0}$);
3. Set the spectral radius of W to α ($W := \alpha \cdot W$).

2.1.2. The training of the ESN

Since the ESN prediction model's parameter is an initialization parameter that does not need to be modified except for the reserve pool and output layer connection matrix, we only need to train W_{out}. Let's say we have a total of training data, let $U = (u(1), u(2), \ldots, u(t)), X = (x(1), x(2), \ldots, x(t)), Y = (y(1), y(2), \ldots, y(t))$, where $u(i), x(i), y(i)$ are the input layer data, reserve pool data and output layer data of the i training set respectively. According to the definition above, we can set $X = S^T \cdot U, Y = D^T \cdot U$. The common training method are as follows:

Pseudo inverse method

$$W_{out} = (S^{\dagger}D)^T \tag{2}$$

where $S^{\dagger}$ is the pseudo inverse of S.

Wiener-Hopf method

$$W_{out} = ((S^TS)^{-1}S^TD)^T \tag{3}$$

Ridge regression(Tikhonov regularization)

$$W_{out} = ((S^TS + \alpha^2 I)^{-1}S^TD)^T \tag{4}$$

When completing the training of the ESN model and need to predict the data, we can get the prediction result only according to the iteration Eq. (1). Its flowchart is shown in Fig. 2.

2.2. Compressed sensing

2.2.1. The basic idea of compressed sensing

According to the Nyquist sampling theorem, in order for the sampled digital signal to retain all the information in the original signal, the sampling frequency must be more than 2 times of the highest frequency in the signal. Compressed sensing believes that if the signal is sparse, it can be reconstructed and restored by sampling points far below the requirements of the sampling theorem. The key to the breakthrough is the sampling mode, and when we say "sampling frequency", that means sampling at equal intervals, which is what digital signals usually do, and that also obeys the Nyquist sampling theorem. If random subsampling is adopted, then the frequency domain will no longer be extended with a fixed period, but will produce a large number of irrelevant disturbance values. The largest peaks are still faintly visible, and these interference values are somewhat overwritten, which look like random noise and are actually caused by the non-zero energy leakage of several original signals. This can be understood as the random sampling makes the spectrum no longer move in a neat way, but move randomly in a small part. The frequency leakage is uniformly distributed in the whole frequency domain, so the leakage value is relatively small, which makes the recovery possible.

The preconditions of compressed sensing are as follows: 1) The signal is sparse in the frequency domain. The signal needs to be sparse in a transform domain, but usually the signal will not be completely sparse . In fact, as long as it approximately meets the sparsity, in other words, most values tend to zero and only a small number of large non-zero values, it can be considered as a compressible signal, and CS subsampling can be carried out on it. If it is not sparse in the frequency domain, we can do discrete wavelet transform (DWT), discrete cosine transform (DCT) and so on to find its sparse transform. 2) Random subsampling. It has been pointed out above that in the case of random subsampling, the frequency

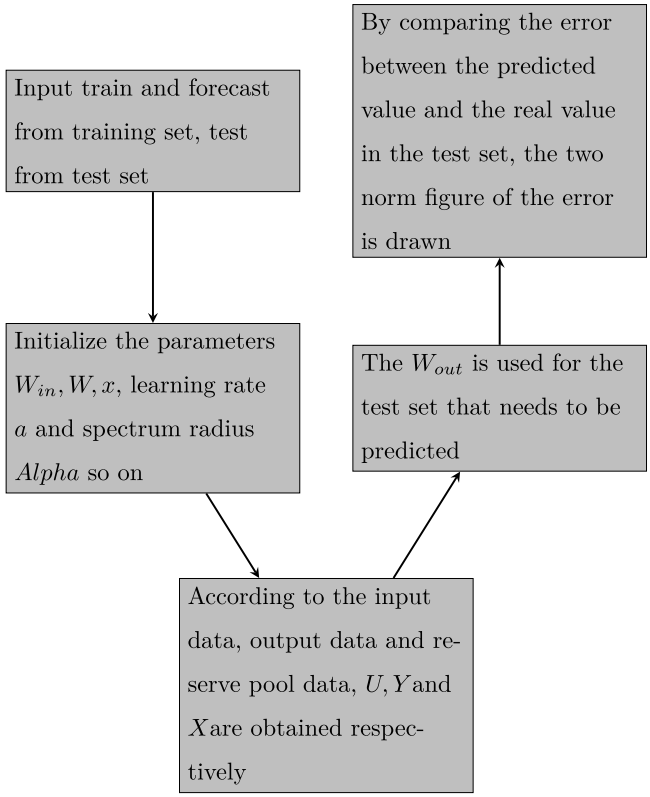


Fig. 2. The flow chart of ESN.

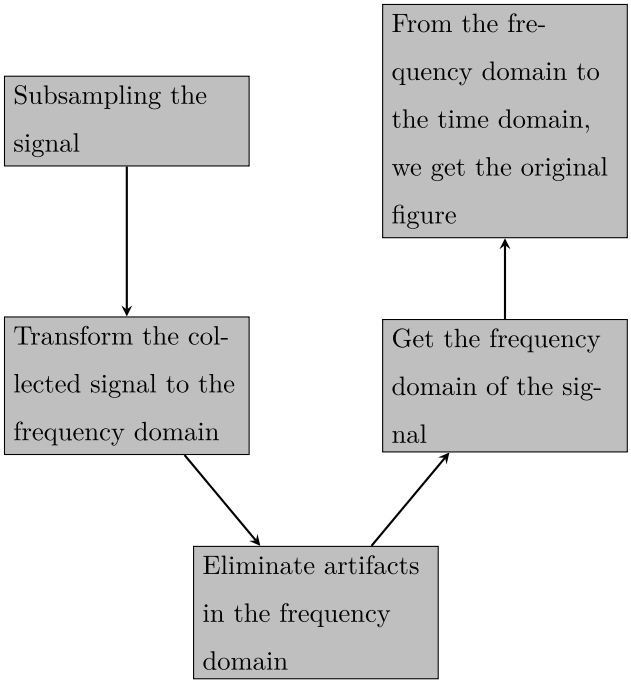


Fig. 3. Compressed sensing flow chart.

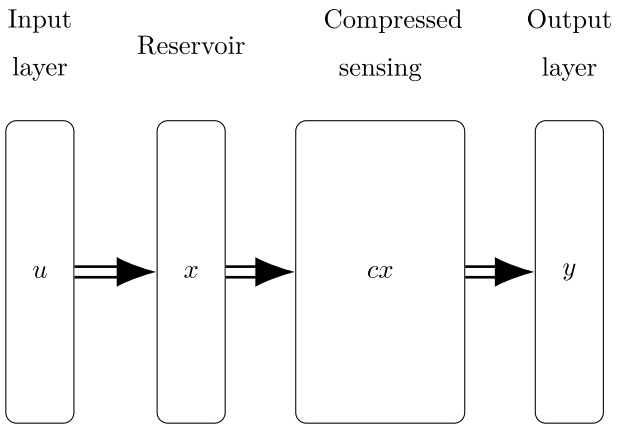


Fig. 4. The ESN model with CS.

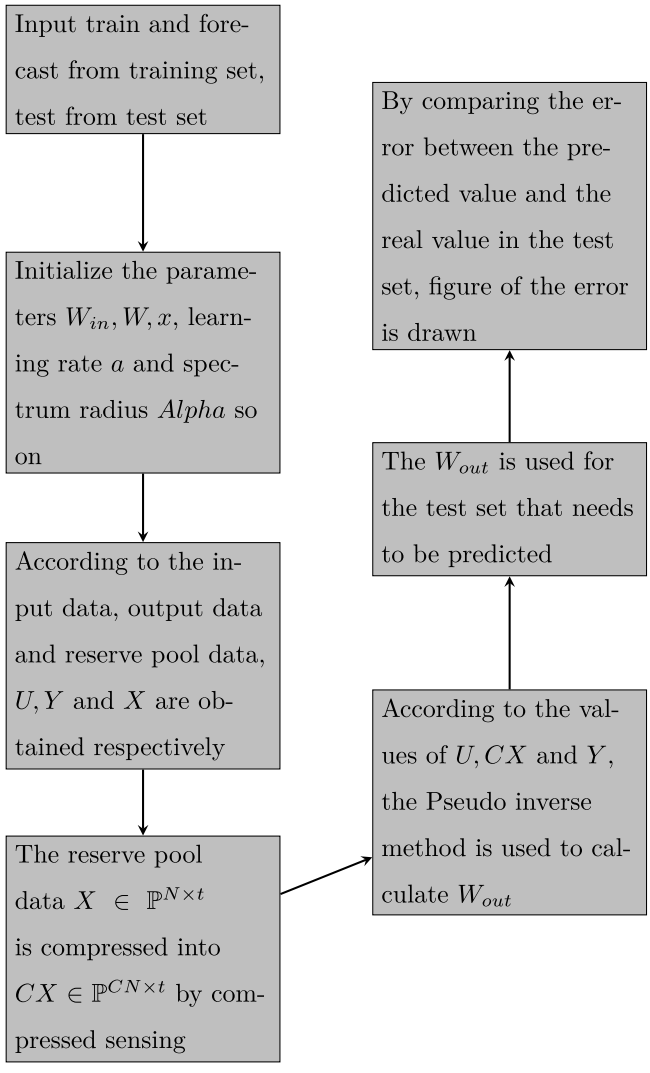


Fig. 5. The ESN flowchart of CS.

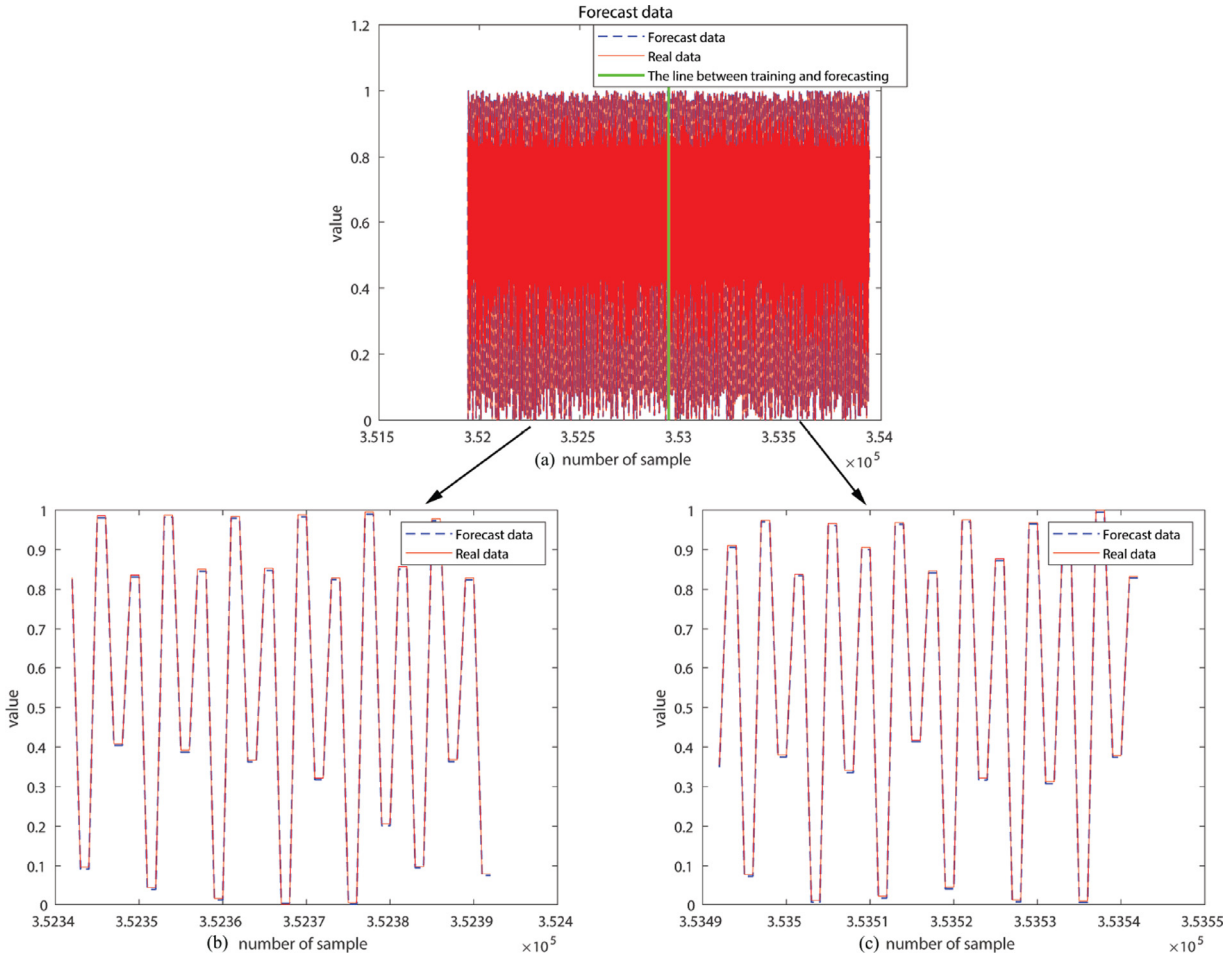


Fig. 6. Comparison between real data and ESN predicted data (part), where both (b) and (c) are part sub-figures of (a).

leakage is uniformly distributed in the whole frequency domain, so the leakage value is relatively small, which makes it possible to recover. Fig. 3 gives the flowchart of compressed sensing.

The Eqs. (5) and (6) can be expressed as follows.

$$y = \Phi \cdot x \tag{5}$$

$$x = \Psi \cdot s \tag{6}$$

Substitute the Eq. (6) into the Eq. (5) to get the Eq. (7),

$$y = \Phi\Psi \cdot s \tag{7}$$

We let $\Theta = \Phi\Psi$ and obtain,

$$y = \Theta \cdot s \tag{8}$$

The Eq. (8) is the mathematical representation of compressed sensing. Where s represents the signal, Ψ represents the sparse basis, x represents the sparse signal, Φ represents the observation matrix, y represents the observed value and $\Theta = \Phi\Psi$.

s is an unknown original signal vector with length N, Φ is the observation matrix, which corresponds to the subsampling process. It is known that Φ projects the high-dimensional signal s to the low-dimensional vector. $y = \Theta \cdot s$ is the observation value with length M, in other words, the result after subsampling, and it's obviously known as well. Therefore, the compressed sensing problem is to solve the underdetermined equations $y = \Theta \cdot s$ to get the original signal s on the basis of known measurement value y and $\Theta = \Phi \cdot \Psi$. However, the general signal s itself is not sparse and needs to be represented sparsely on some sparse basis. Let $x = \Psi \cdot s$, Ψ be the sparse basis matrix and x be the sparse signal. So the final equation becomes: $y = \Phi\Psi \cdot s$, given y, Φ, Ψ solve s.

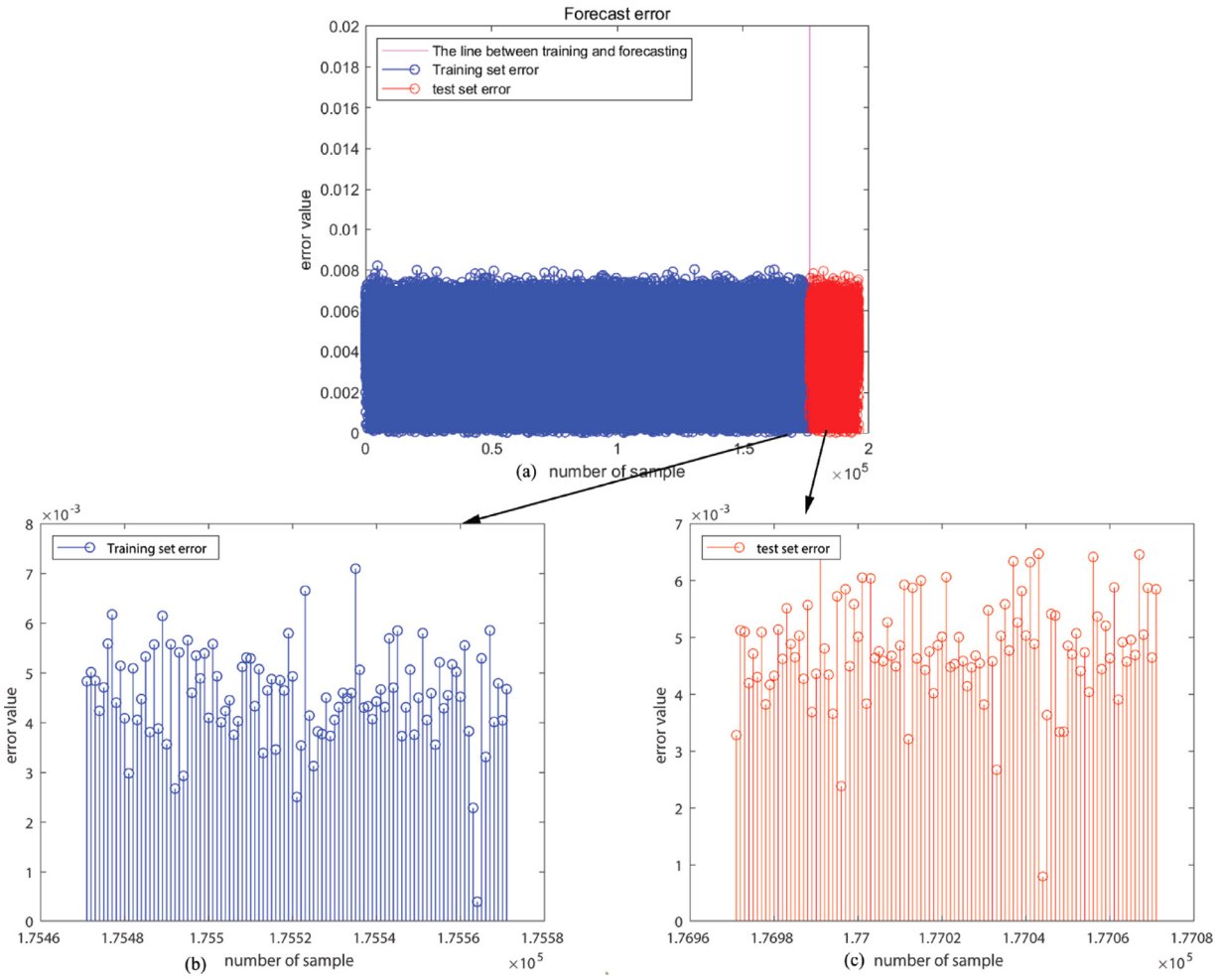


Fig. 7. Error between real data and ESN prediction data, where both (b) and (c) are part sub-figures of (a).

To achieve signal recovery by using random subsampling, Tao and *Candès* proposed in 2005 that the observation matrix Φ should satisfy the condition of Restricted Isometry Property (RIP) [15]:

For $\forall c \in \mathbb{R}^{|T|}$ and constant $\delta_k \in (0,1)$, if:

$$(1-\delta_k)\cdot\|c\|_2^2 \le \|\tilde{\Phi}_T \cdot c\|_2^2 \le (1+\delta_k)\cdot\|c\|_2^2 \tag{9}$$

was established, where $T \subset \{1,2,\ldots,N\}$, $|T| \le k$, $\tilde{\Phi}_T$ is a submatrix whose size is $K \times |T|$formed by the related columns indicated by the index T in $\tilde{\Phi}$, then the matrix $\tilde{\Phi}$ satisfies the Restricted Isometry Property.

Baraniuk proved that the equivalent condition of RIP is that the observation matrix and sparse representation basis are not relevance [16], in other words, $y = \Phi\Psi \cdot s$ in Φ and Ψ are not correlated. Where the relevance is defined as: $u(\Phi,\Psi) = \sqrt{n}\cdot \max_{1\le k,j\le n} |<\Phi_k, \Psi_j>|$, The range of u is $u(\Phi,\Psi) \in [1,\sqrt{n}]$, The smaller u is, the less Φ and Ψ are related.

Tao and *Candès* proved that an independent identically distributed Gaussian random measurement matrix can be a generalized compressed sensing measurement matrix.

2.2.2. Reconstruction of compressed sensing

If a signal is sparse in a transformation domain, then an observation matrix unrelated to the transformation basis can be used to project the higher dimension obtained from the transformation to a lower dimension space, and then the original signal can be reconstructed from the small number of projected values with high probability by solving an optimization problem [17,18]. Reconstruction of CS is the method to solve underdetermined equations $y = \Theta \cdot s$. This is a zero norm (L_0) minimization problem, and an NP complete problem (no quick solution problem), therefore tends to be converted to a norm (L_1) minimization solution, or some approximate estimation algorithms are used [8,19,20].

Orthogonal Matching Pursuit algorithm is one of the commonly used algorithms in CS reconstruction algorithm. Its main ideas are as follows:

1) Since the frequency non-zero value of the original signal still retains a large value in the frequency domain after subsampling, the larger value can be detected by setting a threshold value.

2) Assuming that only these K values exist in the signal, the interference caused by these K values can be calculated.

3) A new non-zero value can be detected by subtracting these K non-zero values and the interference caused by these K values from the original spectrum diagram, and then setting the threshold.

4) If there are more non-zero values in the original signal frequency domain, they can be solved by iteration.

Therefore, we can get the following Algorithm 1.

Algorithm 1 Orthogonal Matching Pursuit.

Input:

Measurement Matrix, Φ;
Sparse Matrix, Ψ;
Observed Signal, y;

Output:

Original Signal, s;

1: Let A be equal to the measurement matrix $A = \Phi$;
2: Let b_i be the i column data of A, where $i = 1, 2, \ldots$;
3: Standardize the b_i, in other words, $\hat{b}_i = \frac{b_i}{\|b_i\|}$, and each normalized sparse basis vector is formed into a new matrix $\hat{A} = [\hat{b}_1, \hat{b}_2, \ldots]$;
4: The residuals of the observations start out as the observations themselves, in other words, $r = y$;
5: **repeat**
6: Calculate the correlation of each standardized sparse basis vector $\hat{b}_i$ with the observed signal residual r, in other words, $w = \hat{A}^T \cdot r$;
7: Find the sparse basis vector with the highest correlation $\hat{b}_{sn}$, delete this vector from $\hat{A}$, and add to the last column of the matrix A_{new}, in other words, $A_{new} = [\hat{b}_{s1}, \hat{b}_{s2}, \ldots, \hat{b}_{sn}]$;
8: Calculate A_{new} pseudo inverse and the product of y have $L_p = A_{new}^{\dagger} \cdot y$, get coefficient vector $X_{rec} \in \mathbb{R}^{N\times 1}$ according to L_p, where N is the length of the original signal, the value of the $s1, s2, \ldots, sn$ row of X_{rec} is the value of the $1, 2, \ldots, n$ of L_p, the remaining rows of X_{rec} have values of 0;
9: Calculate the residual for the remainder $r = y - A_{new} \cdot L_p$;
10: **until** The value of all non-zero values of sparse signal x or residual r is small enough
11: X_{rec} is the signal that is sparse, in other words, $x = X_{rec}$. Calculated original signal $s = \Psi^{-1} \cdot x$;
12: **return** s;

3. ESN model with compressed sensing

The nodes of the reserve pool in ESN model are very many, so when it comes to the operation of the reserve pool, both the operation speed and the accuracy of the operation will definitely be seriously affected. Because the reserve pool node number directly affects how well the ESN model to predict nonlinear data, we can't reduce the reserve pool of nodes directly. Compressed sensing can extract node features, so you can use compressed sensing to reduce the number of nodes in the reserve pool. Due to a small number of nodes contain all the features of the original node, which can be used to describe the original node. In fact, this process is the recovery process of compressed sensing. The nodes extracted from the ESN model can not only ensure the accuracy of the original model, but also enhance its robustness and speed up of its prediction.

Only the connection matrix (W_{out}) from the reserve pool to the output layer needs training in the ESN model. If the nodes after feature extraction are used to train ESN model, which will improve the weight distribution of feature nodes during the training process. This makes the model training process more stable. The ESN model with compressed sensing (for short: CS-ESN model) can be represented as Fig. 4. A compression layer was added to the original ESN model, where the inverse process of compressed sensing is used. The characteristics of the original reserve pool node x were extracted to obtain cx, and then cx was used to train the output layer connection matrix W_{out}. The W_{out} trained by this method has stronger stability and faster prediction speed.

We assume that the reserve pool dimension of the original ESN model is N, and the reserve pool dimension after dimension reduction through compressed sensing is CN. In other words, we first get the original reserve pool data x according to compressed sensing and the sparse basis Ψ, the measurement matrix Φ. Then, cx is obtained by compression of x using compressed sensing. That's the Eq. (10).

$$cx = \Phi\Psi \cdot x \tag{10}$$

By compression sensing, $W_{in} \in \mathbb{P}^{CN\times K}$, $W \in \mathbb{P}^{CN\times CN}$, $W^{out} \in \mathbb{P}^{L\times CN}$. The iterative formula is Eq. (11).

$$x^{(n+1)} = f(W_{in} \cdot u^{(n+1)} + W \cdot x^{(n)})$$

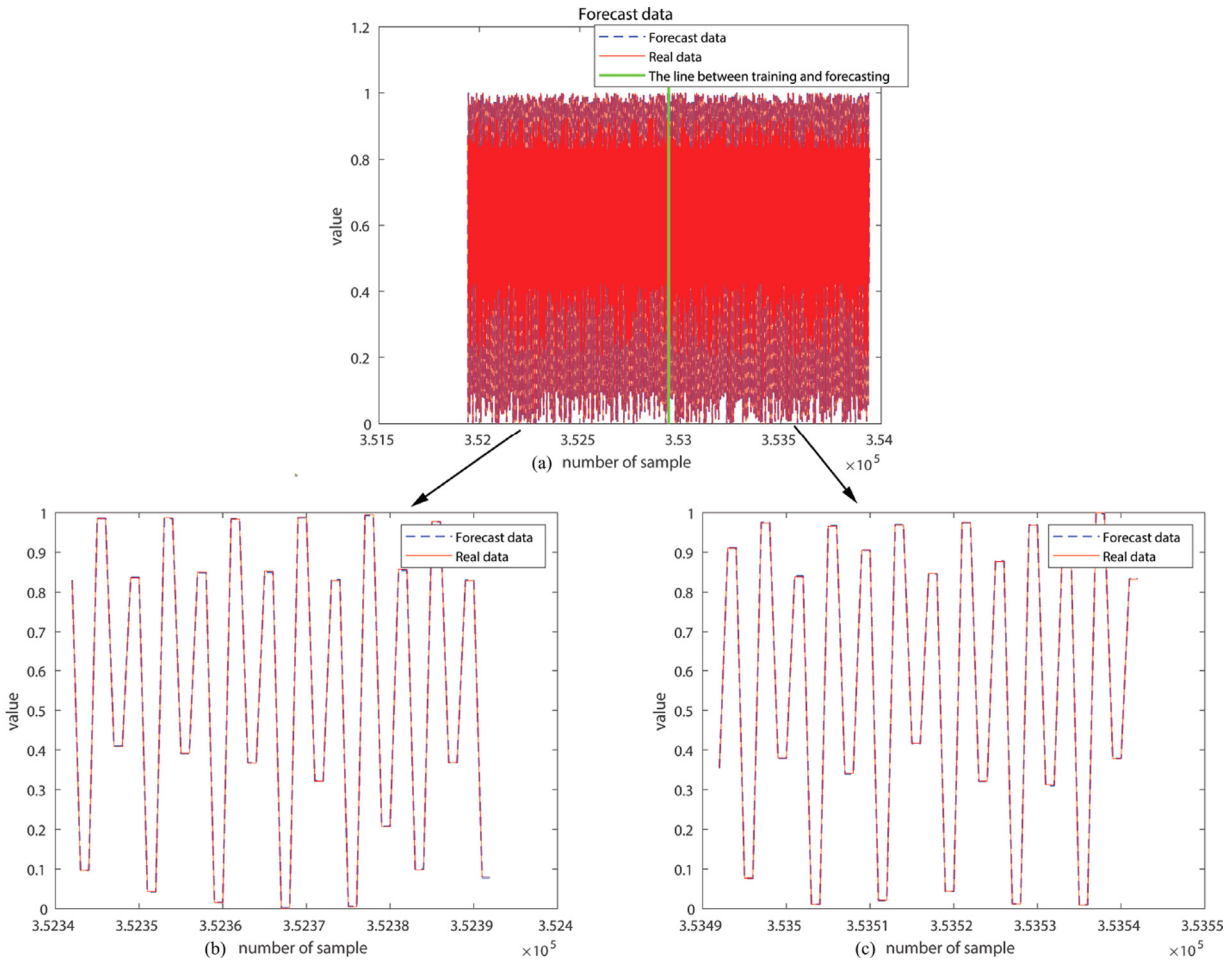


Fig. 8. Comparison between real data and ESN prediction data with compressed sensing (part), where both (b) and (c) are part sub-figures of (a).

$$\begin{aligned} cx^{(n+1)} &= \Theta \cdot x^{(n+1)} = \Phi\Psi \cdot x^{(n+1)} \\ y^{(n+1)} &= f_{out}(W_{out} \cdot cx^{(n+1)}) \end{aligned} \tag{11}$$

According to the above formula, flow chart 5 can be obtained.

4. Numerical experiments

4.1. Logistic chaotic sequence prediction

In this experiment, Logistic chaotic system was used. In order to ensure the chaotic characteristics of the obtained data, the system was first iterated for 10^3 times, and then 10^7 chaotic data were taken. Here, the Logistic parameter is selected as: the u value is 3.58, the initial iteration value is 0.005, and then the data is scaled to the interval [0,1]. The 10^7 data generated by the Logistic chaotic system were divided into 196,078 groups of data, 176,471 groups of training sets, 19,607 groups of test sets, and 5 groups of initialization data. We select the number of input nodes as 50, output nodes as 1, and reserve pool nodes as 2048. The activation function of reserve pool node uses identify function, the learning rate is 0.85, and the spectral radius is 0.25. The ESN model can be trained by Eq. (1), flow chart 2 and Algorithm 1. The comparison diagram between the real data and ESN prediction data is shown in Fig. 6. The error between the real data and ESN prediction data is shown in Fig. 7, where the error calculation formula is $ERROR = |\hat{y}_i - y_i|$, $\hat{y}_i$ is predicted value, and y_i is true value. At the same time, MAE evaluation index is introduced as a comparison between the two models to predict the degree of good or bad, where the MAE calculation formula is $MAE = \frac{1}{n} \cdot \sum_{i=1}^{n} |\hat{y}_i - y_i|$. The evaluation index is also adopted in the next numerical simulation.

In order to test the quality of the ESN model with compressed sensing added, all parameters in the original CS-ESN model remain unchanged, and only the nodes of the reserve pool are compressed, where the compression rate is 50%. Since the data range itself is between [0,1], the sparse basis uses the identity matrix, and the measurement matrix uses part of

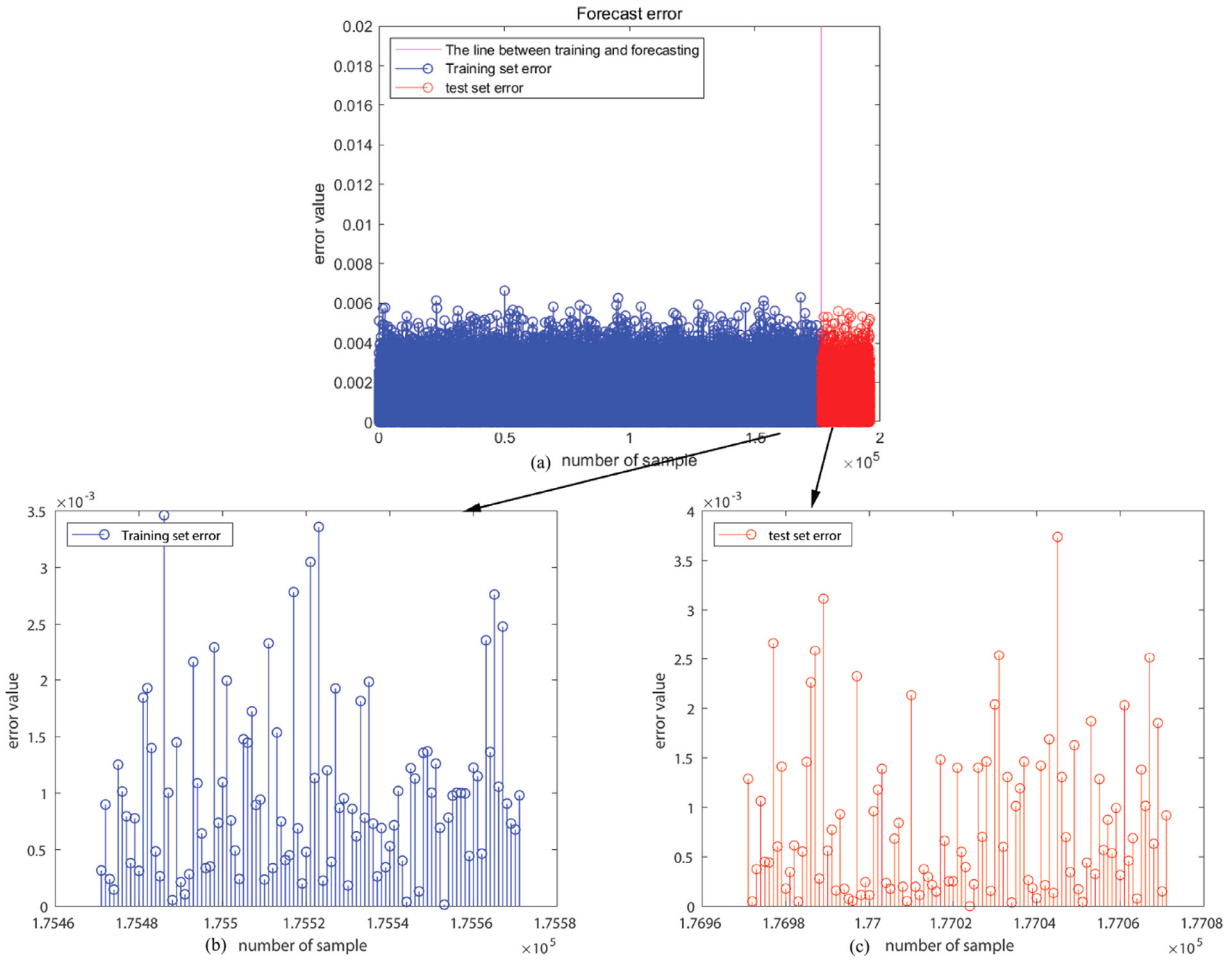


Fig. 9. Error between real data and ESN prediction data with compressed sensing added, where both (b) and (c) are part sub-figures of (a).

Table 2
A comparison diagram of real data and predicted data of chaotic model.

Model/Data	Total sample error	Test set error	MAE of total sample
ESN	980.672523	98.179800	0.005001
CS-ESN	186.676942	18.830397	0.000952
Model/Data	MAE of test set	Total sample variance	Test set variance
ESN	0.005007	0.000001	0.000001
CS-ESN	0.000960	0.000001	0.000001
Model/Data	Samples with an error of more than mean	Samples with an error of more than mean account for the total sample	
ESN	116577	59.45440%	
CS-ESN	78295	39.930538%	

the Hadamard matrix. Since some nodes will exceed the range of [-1,1] after the reserve pool is compressed, it needs to be standardized. The data scale between [-1,1], the processing method is: $cx := cx/(\max(cx) - \min(cx))$. By the compression Eq. (10) and the iteration Eq. (11), the ESN prediction model with CS added can be obtained. The model is trained to compare the error between the real data and the predicted data. Fig. 8 shows the comparison between the real data and the CS-ESN prediction data . Fig. 9 shows the two norms of the error between the real data and the CS-ESN prediction data.

By comparing the original ESN model with the CS-ESN model, it is obvious that the model with compressed sensing is more stable, with smaller error and lower dispersion degree of prediction error (the variance of the prediction error is smaller). The detailed data are shown in Table 2. According to the original ESN error Fig. 7 and the CS-ESN error Fig. 9, the CS-ESN model is better than the original ESN model in both training samples and test samples in addition to the initial sample. Table 2 shows that, 1) The total sample error value and test set error value of the CS-ESN model are better than the original ESN model; 2) The total sample average prediction error and test set average prediction error of the CS-ESN model

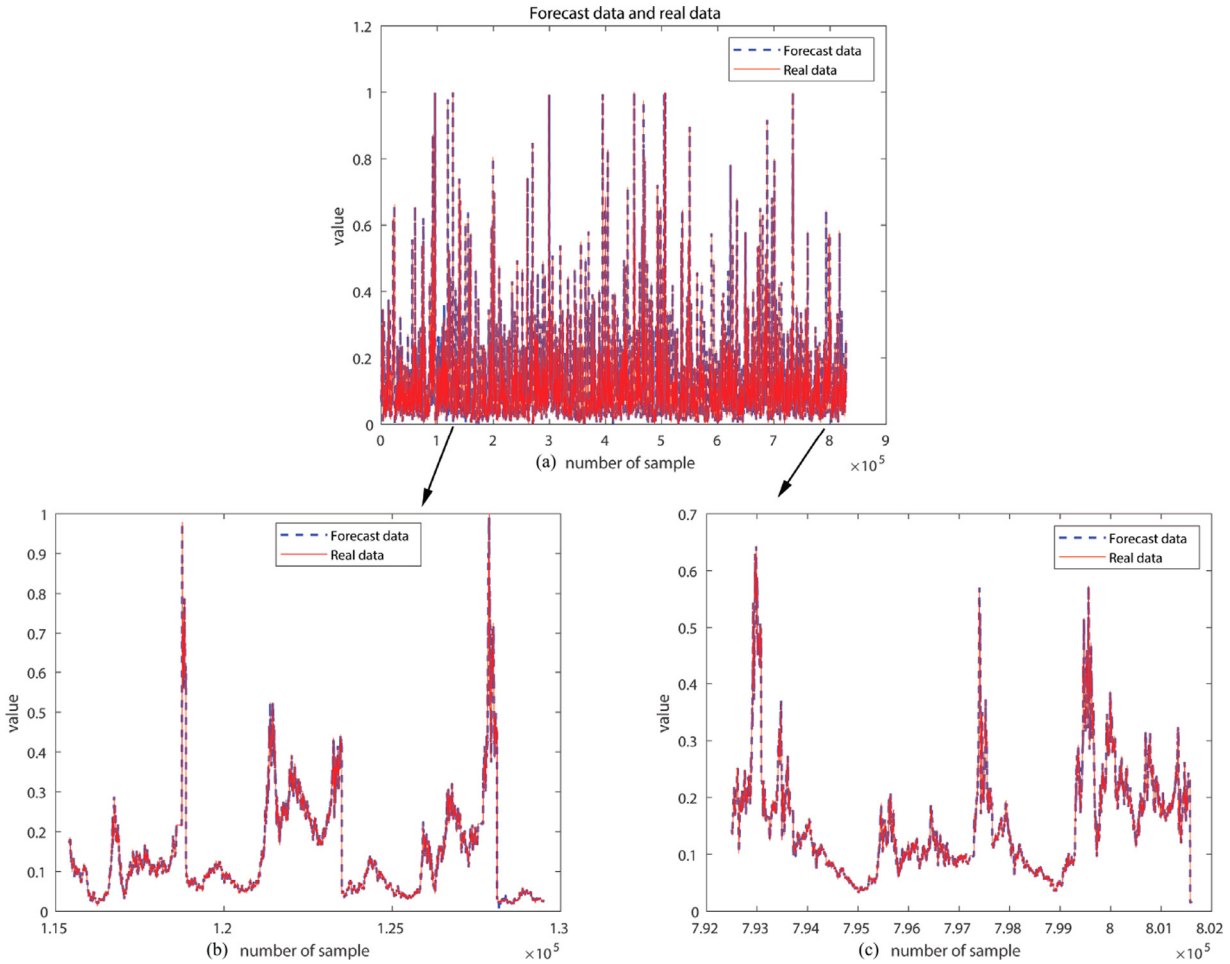


Fig. 10. Comparison of real data and ESN prediction data, where both (b) and (c) are part sub-figures of (a).

are better than the original ESN model, and the accuracy is improved significantly. From Table 2, the MAE of total samples is 0.005001 in the ESN model, and the MAE of test set is 0.005007 in the ESN model. The MAE of total sample is 0.000952 in the CS-ESN model, and the MAE of test set is 0.000960 in the CS-ESN model; 3) The total sample variance and test set variance of the CS-ESN model are less than the original ESN model; 4) The number of samples with an error of more than mean the CS-ESN model is much smaller than the original ESN model, and the proportion of samples with an error of more than 0.003 in the total sample is also significantly reduced. Therefore, the CS-ESN model is greatly improved both in the range of error generation and in the stability of error.

Remark 1. By comparing the prediction result of ESN model 6 and CS-ESN model 8, it can be seen that the true values and predicted values of the CS-ESN model are basically equal, and the true value curve and predicted value curve of the CS-ESN model basically coincide. There is a certain deviation between the true values and the predicted values of the ESN model, and a certain distance between the true value curve and the predicted value curve of the ESN model can be observed. By comparing the error figure of ESN model 7, the error figure of CS-ESN model 9 and Table 2, MAE value of ESN model is 0.005007, and MAE value of CS-ESN model is 0.000960, which indicates that CS-ESN model has better precision on chaotic data than ESN model. At the same time, the variance of prediction error of CS-ESN model and ESN model is basically the same, both of which are 0.000001. This means that CS-ESN model has the same strong robustness as ESN model on chaotic data.

4.2. Stock prediction

This experiment selected the stock data of 200 stocks in the past 17 years (data sources: http://www.sse.com.cn/market/overview/), with a total of 828,799 data, and classified 16,250 sets of data, 14,620 sets of training sets, 1,625 sets of test sets, and 5 sets of initialization data. In order to make the stock data more smooth and stable, the stock data is standardized.

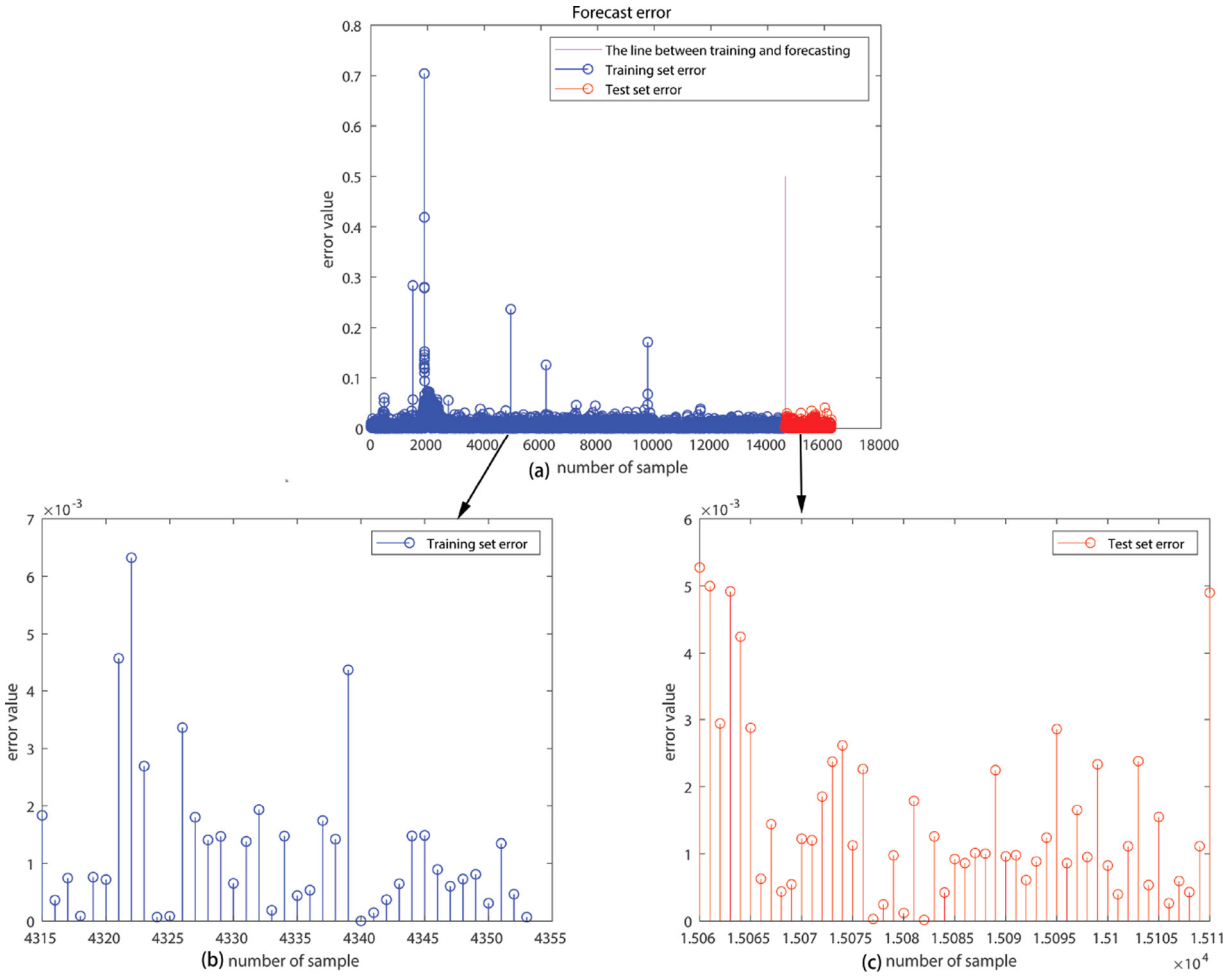


Fig. 11. Error between real data and ESN prediction data, where both (b) and (c) are part sub-figures of (a).

Firstly, the Z-score method is used to normalize the data, that is, the processed data distribution is presented as a normal distribution with a mean of 0 and a variance of 1. The rules of normalization are as follows: the data $x_1, x_2, \ldots, x_n$.

$$\bar{x} = \frac{1}{n} \cdot \sum_{i=1}^{n} x_i$$

$$s = \sqrt{\frac{1}{n-1} \cdot \sum_{i=1}^{n} (x_i - \bar{x})^2}$$

$$y_i = \frac{x_i - \bar{x}}{s}$$

Then, min-max normalization processing is carried out on the data, that is, linear transformation is carried out on the data to make it shrink to the interval of [0,1]. The rules of min-max normalization processing are as follows: the data $x_1, x_2, \ldots, x_n$.

$$y_i = \frac{x_i - \min\limits_{1 \le j \le n} \{x_j\}}{\max\limits_{1 \le j \le n} \{x_j\} - \min\limits_{1 \le j \le n} \{x_j\}}$$

We select the number of input nodes as 50, output nodes as 1, and reserve pool nodes as 2048. The activation function of reserve pool node uses identify function, the learning rate is 0.8, and the spectral radius is 0.75. The ESN model can be trained by the Eq. (1), flow chart 2 and Algorithm 1. Fig. 10 shows the comparison between real data and ESN prediction data. Fig. 11 shows the two norms of error between real data and ESN prediction data.

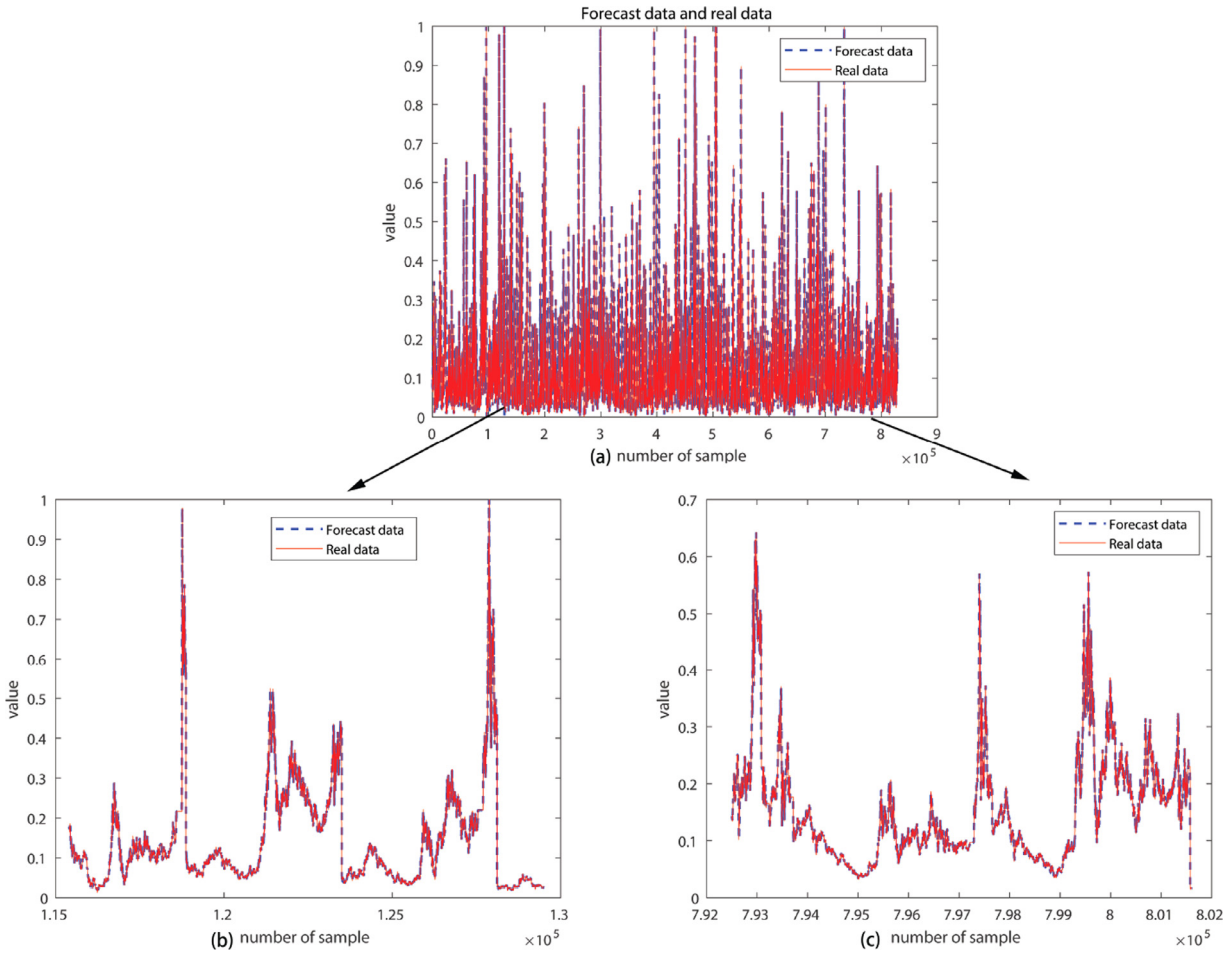


Fig. 12. Comparison of real data and ESN prediction data with compressed sensing added, where both (b) and (c) are part sub-figures of (a).

In order to compare the degree of effect improvement between the CS-ESN model and the original ESN model, all parameters in the original ESN model remain unchanged and only the node of the reserve pool is compressed, where the compression rate is 50%, the discrete wavelet basis is selected for the sparse basis, and part of the Hadamard matrix is selected for the measurement matrix. Since some nodes may overflow after the reserve pool is compressed, all nodes in the reserve pool are scaled down to between [-1, 1] in the same way as in the previous experiment. Through the compression Eq. (10) and the iterative Eq. (11), the CS-ESN prediction model can be obtained, and the model can be trained, then compare the error value between the real data and the predicted data. Fig. 12 shows the comparison between the real data and the ESN prediction data with compressed sensing. Fig. 13 shows the two norms of error between the real data and the CS-ESN prediction data.

At the same time, the total error value and the test set error value of the original ESN model and the CS-ESN model can be obtained, where the total error value refers to the sum of the two norms of the training set and the test set error, and the test set error value refers to the sum of the two norms of the test set error. The total error value of the original ESN model is 66.564432, the test set error value is 4.991433, the MAE of total sample is 0.004096, the MAE of test set is 0.003072, the total sample variance is 0.000122, the test set variance is 0.000016. The total error value of the CS-ESN model is 45.667224, the total error value of the test set is 3.715955, the MAE of total sample is 0.002810, the MAE of test set is 0.002287, the total sample variance is 0.000058, and the variance of the test set is 0.000011. According to Table 3, it is easy to see that the CS-ESN model is significantly better than the original ESN prediction model, and the stability of the CS-ESN model is also better than the original ESN prediction model.

Remark 2. By comparing the forecast figures of ESN model 10 and CS-ESN model 12, both ESN model and CS-ESN model can predict the trend of stocks very well. However, according to the error figure of ESN model 11 and CS-ESN model 13, it can be seen that CS-ESN model is better than ESN model in predicting the accuracy of stock data except for some points. According to the Table 3, the MAE value of the CS-ESN model is 0.0025, while the MAE value of the ESN model is 0.0035, which shows that the CS-ESN model has higher accuracy in stock data than the ESN model. Meanwhile, the prediction

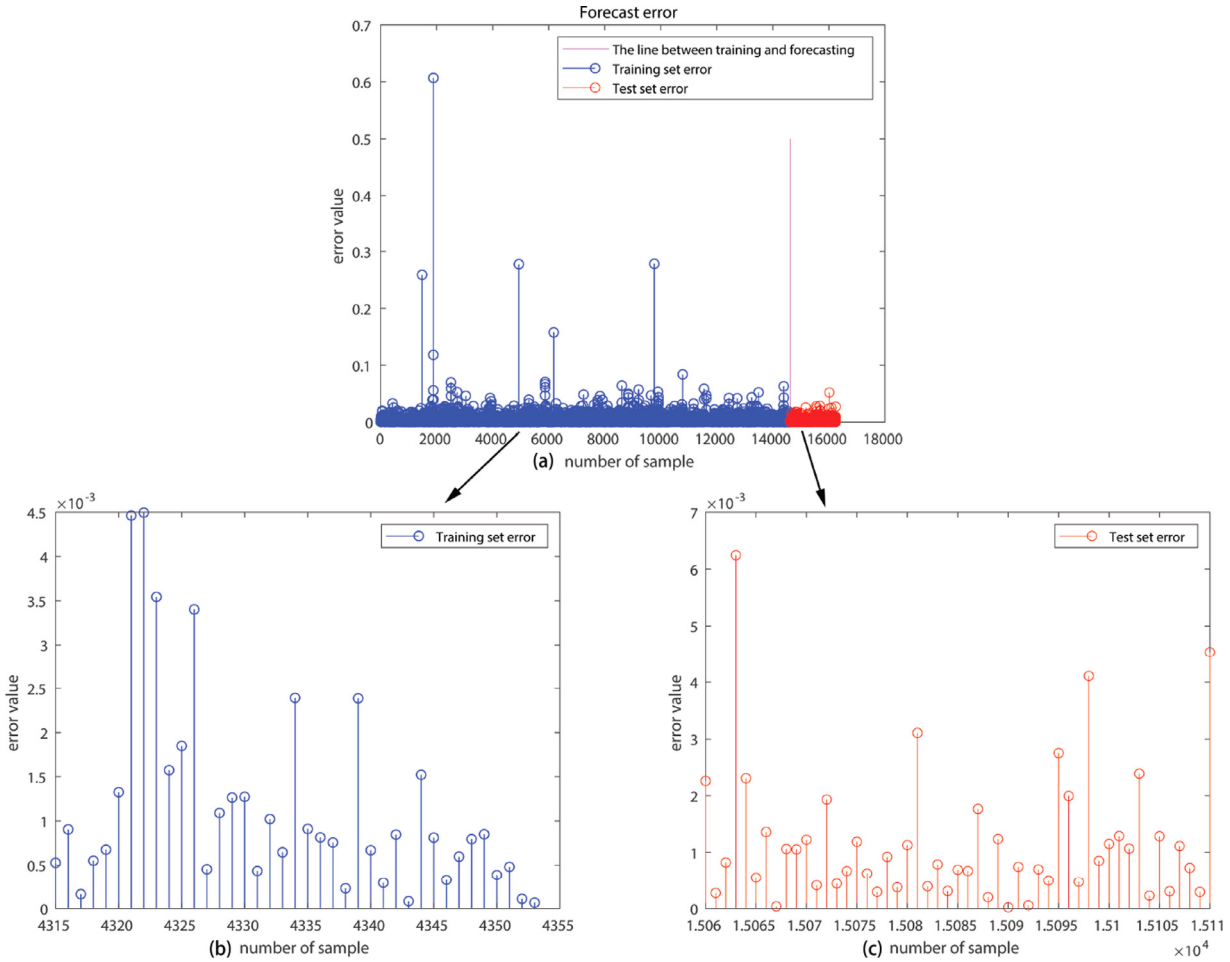


Fig. 13. Error between real data and ESN prediction data with compressed sensing added, where both (b) and (c) are part sub-figures of (a).

Table 3
A chart comparing real stock data with forecast data.

Model/Data	Total sample error	Test set error
ESN	66.564432	4.991433
CS-ESN	45.667224	3.715955
Model/Data	MAE of total sample	MAE of test set
ESN	0.004096	0.003072
CS-ESN	0.002810	0.002287
Model/Data	Total sample variance	Test set variance
ESN	0.000122	0.000016
CS-ESN	0.000058	0.000011

error variance of the CS-ESN model is 0.000030, while the prediction error variance of the ESN model is 0.000069, which indicates that the CS-ESN model has stronger robustness than the ESN model on stock data.

5. Conclusion

In this paper, a new ESN model is proposed, and compressed sensing is added on the original ESN prediction model. The purpose of extracting the characteristics of the reserve pool node is to reduce the reserve pool node. Since the feature nodes of the reserve pool are extracted, most of the features of the original reserve pool are retained. Therefore, the CS-ESN model can ensure the accuracy of prediction and even have a better effect than the original ESN model. It can be seen from the simulation that both the prediction accuracy and the prediction stability have been improved greatly. Through the simulation of chaotic series prediction and stock prediction, both the average value of prediction error and the variance of

prediction error are obviously reduced. Therefore, CS-ESN model has a good effect in chaotic series prediction and has a good application market in stock prediction.

Declaration of Competing Interest

We declare that we do not have any commercial or associative interest that represents a conflict of interest in connection with the work submitted.

CRediT authorship contribution statement

Hao Zhang: Funding acquisition, Writing - original draft. **Mingwen Zheng:** Conceptualization, Methodology, Supervision, Funding acquisition. **Yanping Zhang:** Investigation, Resources. **Xiao Yu:** Investigation. **Wenchao Li:** Writing - original draft, Validation. **Hui Gao:** Writing - original draft.

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